Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)
Solutions to the Exercises of Chapter 9: Lorenz Equations

Conventions. The Lorenz system is

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - xz - y, \qquad \dot{z} = xy - bz,$$

with $\sigma, b, r > 0$. The origin is a fixed point for all r; for $r > 1$ there are two more, $C^{\pm} = (\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1)$. The "standard" values are $\sigma = 10$, $b = 8/3$.

About the numerical exercises. Every numerical exercise below was actually integrated (Runge–Kutta/LSODA with tolerances 10^{-8} or tighter) and the figures are from those runs. Where a conclusion is only supported by finite-time numerics, I say so. Statements taken from the literature rather than from my own runs are marked as such.

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1 A Chaotic Waterwheel

Recall the waterwheel model of Section 9.1. The mass density $m(\theta, t)$ satisfies

$$\frac{\partial m}{\partial t} = Q - Km - \omega\frac{\partial m}{\partial \theta}, \qquad I\dot{\omega} = -\nu\omega + gr\int_0^{2\pi} m\sin\theta\,\mathrm{d}\theta, \tag{1}$$

and with $m = \sum_n [a_n \sin n\theta + b_n \cos n\theta]$, $Q = \sum_n q_n \cos n\theta$:

$$\dot{a}_n = n\omega b_n - K a_n, \tag{2}$$

$$\dot{b}_n = -n\omega a_n - K b_n + q_n, \tag{3}$$

$$\dot{a}_1 = \omega b_1 - K a_1, \quad \dot{b}_1 = -\omega a_1 - K b_1 + q_1, \quad \dot{\omega} = \frac{-\nu\omega + \pi g r\, a_1}{I}. \tag{4}$$

9.1.1 *(Waterwheel's moment of inertia approaches a constant)*

Show that $I(t) \to$ const. as $t \to \infty$.

Solution. **(a)** Water at angle θ sits on the rim, at distance r from the axis, so a mass $m\,\mathrm{d}\theta$ contributes $r^2 m\,\mathrm{d}\theta$ to the moment of inertia:

$$I_{\text{water}} = \int_0^{2\pi} r^2\, m(\theta, t)\,\mathrm{d}\theta = r^2 M, \qquad I = I_{\text{wheel}} + r^2 M.$$

(b) Differentiate $M = \int_0^{2\pi} m\,\mathrm{d}\theta$ and use (1):

$$\dot{M} = \int_0^{2\pi} \Big(Q - Km - \omega \frac{\partial m}{\partial \theta}\Big)\mathrm{d}\theta = Q_{\text{total}} - KM - \omega\big[m(2\pi, t) - m(0, t)\big] = Q_{\text{total}} - KM,$$

because m is 2π-periodic in θ.
(c) This linear equation has solution $M(t) = \frac{Q_{\text{total}}}{K} + \big(M(0) - \frac{Q_{\text{total}}}{K}\big)e^{-Kt}$, so $M \to Q_{\text{total}}/K$ and

$$I(t) \;\longrightarrow\; I_{\text{wheel}} + \frac{r^2 Q_{\text{total}}}{K}.$$

■

9.1.2 *(Behavior of higher modes)*

Solution. **(a)** If $Q = q_1 \cos\theta$ then $q_n = 0$ for $n \neq 1$. For such n, (2)–(3) give

$$\frac{\mathrm{d}}{\mathrm{d}t}(a_n^2 + b_n^2) = 2a_n(n\omega b_n - K a_n) + 2b_n(-n\omega a_n - K b_n) = -2K(a_n^2 + b_n^2).$$

The ω-terms cancel identically, so regardless of what $\omega(t)$ does, $a_n^2 + b_n^2 = (a_n^2 + b_n^2)(0)\, e^{-2Kt} \to 0$. (For $n = 0$ the a_0 term is absent and $\dot{b}_0 = -K b_0$.)

(b) For general Q, put $z_n = a_n + i b_n$. Then (2)–(3) read

$$\dot{z}_n = -(K + in\omega(t))\, z_n + i q_n \qquad (n \neq 1).$$

Hence $\frac{\mathrm{d}}{\mathrm{d}t}|z_n|^2 = -2K|z_n|^2 + 2q_n b_n \leq -2K|z_n|^2 + 2|q_n||z_n|$, i.e. $\dot{|z_n|} \leq -K|z_n| + |q_n|$, so $\limsup_{t\to\infty} |z_n| \leq |q_n|/K$: the higher modes stay bounded but need not decay. The homogeneous part decays like e^{-Kt}, so after transients

$$z_n(t) = i q_n \int_0^{\infty} e^{-Ks} \exp\Big(-in\int_{t-s}^{t} \omega(u)\,\mathrm{d}u\Big)\mathrm{d}s,$$

i.e. each higher mode is *slaved* to the history of $\omega(t)$. If $\omega \to \omega^*$ (a steady fixed point), then $z_n \to i q_n/(K + in\omega^*)$. Since a_n, b_n $(n \neq 1)$ never feed back into (a_1, b_1, ω), they do not affect

whether the three-variable dynamics is chaotic; they just ride along on it. (This is as far as the elementary argument goes; the full state of knowledge is in Kolar & Gumbs 1992.) ■

9.1.3 *(Deriving the Lorenz equations from the waterwheel)*

Solution. Let $\tau = Kt$ (primes denote $\mathrm{d}/\mathrm{d}\tau$) and set

$$\omega = Kx, \qquad a_1 = \beta y, \qquad b_1 = \frac{q_1}{K} - \beta z, \qquad \beta = \frac{\nu K}{\pi g r}.$$

Then:

$$\begin{aligned} x' &= \frac{1}{K^2}\dot{\omega} = \frac{1}{K^2}\cdot\frac{-\nu K x + \pi g r \beta y}{I} = \frac{\nu}{IK}(y - x), \\ y' &= \frac{1}{K\beta}\dot{a}_1 = \frac{1}{K\beta}\big(Kx(\tfrac{q_1}{K} - \beta z) - K\beta y\big) = \frac{q_1}{K\beta}x - xz - y, \\ z' &= -\frac{1}{K\beta}\dot{b}_1 = \frac{1}{K\beta}\big(\omega a_1 + K b_1 - q_1\big) = \frac{1}{K\beta}\big(K\beta xy - K\beta z\big) = xy - z. \end{aligned}$$

This is the Lorenz system with

$$\boxed{\sigma = \frac{\nu}{IK}, \qquad r = \frac{q_1}{K\beta} = \frac{\pi g r_{\mathrm{w}}\, q_1}{K^2 \nu}, \qquad b = 1}$$

(r_{w} = wheel radius, to avoid confusion with the Lorenz r). So $b = 1$: the waterwheel is a special case. In the usual language, σ is the Prandtl number and r the Rayleigh number. ■

9.1.4 *(Laser model)*

$\dot{E} = \kappa(P - E)$, $\dot{P} = \gamma_1(ED - P)$, $\dot{D} = \gamma_2(\lambda + 1 - D - \lambda EP)$.

Solution. **(a)** If $E^* = 0$ then $P^* = 0$ and $D^* = \lambda + 1$. Linearizing about $(0, 0, \lambda + 1)$ (the λEP term is second order):

$$\delta\dot{D} = -\gamma_2\,\delta D, \qquad \begin{pmatrix} \delta\dot{E} \\ \delta\dot{P} \end{pmatrix} = \begin{pmatrix} -\kappa & \kappa \\ \gamma_1(\lambda+1) & -\gamma_1 \end{pmatrix}\begin{pmatrix} \delta E \\ \delta P \end{pmatrix}.$$

The trace is $-(\kappa + \gamma_1) < 0$ and the determinant is $-\kappa\gamma_1\lambda$. So the non-lasing state is stable for $\lambda < 0$ and a saddle for $\lambda > 0$: the **laser threshold is** $\lambda_c = 0$. One eigenvalue crosses 0, and the system is symmetric under $(E, P) \to (-E, -P)$, so the bifurcation is a **pitchfork**; it is supercritical (it is the Lorenz pitchfork at $r = 1$, see (b)). For $\lambda > 0$ the lasing states are $E = P = \pm 1$, $D = 1$. (The amplitude ± 1 does not shrink to 0 at threshold only because E is normalized with a λ-dependent factor; see the scaling below.)

(b) For $\lambda > 0$ let $\tau = \gamma_1 t$ and

$$x = \sqrt{b\lambda}\,E, \quad y = \sqrt{b\lambda}\,P, \quad z = (\lambda + 1) - D, \qquad \sigma = \frac{\kappa}{\gamma_1}, \quad b = \frac{\gamma_2}{\gamma_1}, \quad r = \lambda + 1.$$

Then $x' = \sigma(y - x)$; $y' = \sqrt{b\lambda}(ED - P) = x(r - z) - y$; and $z' = -D' = -b(\lambda + 1 - D) + b\lambda EP = -bz + xy$, since $xy = b\lambda EP$. So the laser equations *are* the Lorenz equations, with threshold $\lambda = 0 \leftrightarrow r = 1$. Lasing states $E = \pm 1$ correspond to $C^{\pm}$ ($x^2 = b(r - 1) = b\lambda$). ■

9.1.5 *(Research project: asymmetric waterwheel)*

Solution. This is open-ended; here is how far a short analysis goes. Allow $Q(\theta) = \sum_n [p_n \sin n\theta + q_n \cos n\theta]$. Substituting in (1) gives

$$\dot{a}_n = n\omega b_n - K a_n + p_n, \qquad \dot{b}_n = -n\omega a_n - K b_n + q_n.$$

The torque equation is unchanged (only a_1 survives the $\sin\theta$ integral), so (a_1, b_1, ω) is still a closed system, now with a new constant term:

$$\dot{a}_1 = \omega b_1 - K a_1 + \mathbf{p_1}, \qquad \dot{b}_1 = -\omega a_1 - K b_1 + q_1, \qquad I\dot{\omega} = -\nu\omega + \pi g r\, a_1. \tag{5}$$

In Lorenz form (same scalings as 9.1.3): $x' = \sigma(y - x)$, $y' = rx - xz - y + h$, $z' = xy - z$ with $h = p_1/(K\beta)$. The system still has constant divergence $-(2K + \nu/I)$, so volume contraction, and all the trapping-region arguments of Section 9.2 survive. The symmetry $(x, y, z) \to (-x, -y, z)$ is broken when $h \neq 0$.

Fixed points. Put $s = \nu/(\pi g r)$, so $a_1 = s\omega$. From the $\dot{b}_1$ and $\dot{a}_1$ equations, $b_1 = (Ks\omega - p_1)/\omega$ and $Kb_1 = q_1 - s\omega^2$, hence

$$s\omega^3 - (q_1 - K^2 s)\,\omega - K p_1 = 0 \quad \Longleftrightarrow \quad \omega^3 - K^2(r-1)\,\omega - \frac{K p_1}{s} = 0.$$

For $p_1 = 0$ this is the pitchfork $\omega(\omega^2 - K^2(r-1)) = 0$. For $p_1 \neq 0$ it is the unfolded cubic of Section 3.6: $\omega = 0$ is no longer a solution, there is a single fixed point for $r - 1 < \left(27 p_1^2/(4K^4 s^2)\right)^{1/3}$ (roughly), and three fixed points beyond that (saddle-node pair appears): the pitchfork is replaced by an **imperfect bifurcation**. A full study of the Hopf bifurcations and chaos for $h \neq 0$ would be the real research project; I have not done that. ■

2 Simple Properties of the Lorenz Equations

9.2.1 *(Parameter where Hopf bifurcation occurs)*

Solution. **(a)** At $C^{\pm}$ we have $r - z^* = 1$, $y^* = x^*$, $x^{*2} = b(r-1)$, and

$$J = \begin{pmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & -x^* \\ x^* & x^* & -b \end{pmatrix}.$$

Expanding $\det(J - \lambda I)$ along the first row:

$$\begin{aligned} \det(J - \lambda I) &= -(\sigma + \lambda)\big[(1+\lambda)(b+\lambda) + x^{*2}\big] - \sigma\big[-(b+\lambda) + x^{*2}\big] \\ &= -\big[\lambda^3 + (\sigma + b + 1)\lambda^2 + b(r + \sigma)\lambda + 2b\sigma(r-1)\big], \end{aligned}$$

using $(1+\lambda)(b+\lambda) + x^{*2} = \lambda^2 + (b+1)\lambda + br$. Hence

$$\lambda^3 + (\sigma + b + 1)\lambda^2 + (r + \sigma)b\lambda + 2b\sigma(r-1) = 0.$$

(b) Put $\lambda = i\omega$, $\omega \in \mathbb{R}$. The real and imaginary parts are

$$-(\sigma + b + 1)\omega^2 + 2b\sigma(r-1) = 0, \qquad -\omega^3 + b(r+\sigma)\omega = 0.$$

For $\omega \neq 0$: $\omega^2 = b(r+\sigma)$ and $\omega^2 = \frac{2b\sigma(r-1)}{\sigma+b+1}$. Equating, $(r+\sigma)(\sigma+b+1) = 2\sigma(r-1)$, i.e. $r(\sigma-b-1) = \sigma(\sigma+b+3)$:

$$\boxed{r_H = \frac{\sigma(\sigma+b+3)}{\sigma-b-1}}, \qquad \omega^2 = b(r_H+\sigma) = \frac{2b\sigma(\sigma+1)}{\sigma-b-1}.$$

We need $\omega^2 > 0$, which (for $\sigma > 0$) requires $\sigma > b+1$. If $\sigma \leq b+1$ there is no Hopf bifurcation for $r > 0$: indeed the Routh–Hurwitz condition $(\sigma+b+1)b(r+\sigma) > 2b\sigma(r-1)$ becomes $r(b+1-\sigma)+\sigma(\sigma+b+3) > 0$, which holds for all $r > 0$, so $C^{\pm}$ stay stable.

(c) The eigenvalues sum to the trace of J which is $-(\sigma+b+1)$. With the other two being $\pm i\omega$,

$$\lambda_3 = -(\sigma+b+1).$$

■

9.2.2 *(An ellipsoidal trapping region)*

Solution. Let $V = rx^2 + \sigma y^2 + \sigma(z-2r)^2$. Then

$$\begin{aligned}\tfrac{1}{2}\dot{V} &= rx\,\sigma(y-x) + \sigma y(rx - xz - y) + \sigma(z-2r)(xy - bz) \\ &= -r\sigma x^2 - \sigma y^2 - \sigma b z^2 + 2r\sigma bz\end{aligned}$$

(the xy and xyz terms cancel). Completing the square in z,

$$\dot{V} = -2\sigma\Big[rx^2 + y^2 + b(z-r)^2 - br^2\Big].$$

So $\dot{V} < 0$ outside the ellipsoid $E_0: \; rx^2 + y^2 + b(z-r)^2 = br^2$. Pick C larger than $\max_{E_0} V$. Then on $\{V \geq C\}$, which is closed and lies strictly outside E_0, the continuous function $f = rx^2 + y^2 + b(z-r)^2 - br^2$ is positive and $\to \infty$ at infinity, hence $f \geq m > 0$ and $\dot{V} \leq -2\sigma m$. Every trajectory therefore reaches $\{V \leq C\}$ in finite time, and cannot leave it since $\dot{V} < 0$ on $V = C$. This is the ellipsoid E.

Smallest C. The boundary must be crossed inward everywhere, i.e. $\{V = C\} \cap \{f \leq 0\} = \emptyset$, so $C_{\min} = \max_{E_0} V$ (a convex function on a convex set, so the maximum is on ∂E_0). Writing $u = z - r$ and eliminating x^2 via the constraint, $V = br^2 + (\sigma-1)y^2 + \sigma(u-r)^2 - bu^2$ with $y^2 \leq b(r^2-u^2)$. For $\sigma \geq 1$ the maximum is at $x = 0$, giving $V = \sigma\big[(b+1)r^2 - 2ru - (b-1)u^2\big]$, $u \in [-r, r]$, maximized at $u = -r/(b-1)$ when $b \geq 2$:

$$C_{\min} = \frac{\sigma b^2 r^2}{b-1} \quad (\sigma \geq 1,\ b \geq 2), \qquad C_{\min} = 4\sigma r^2 \ \ (\sigma \geq 1,\ b \leq 2),$$

(e.g. $\sigma = 10$, $b = 8/3$: $C_{\min} = \frac{\sigma b^2}{b-1} r^2 = \frac{128}{3} r^2 \approx 42.7\, r^2$). For $\sigma < 1$ the analogous computation gives $r^2 b^2/(b-\sigma)$ if $b \geq 2\sigma$ and $4\sigma r^2$ otherwise. ■

9.2.3 *(A spherical trapping region)*

Solution. Let $W = x^2 + y^2 + (z-r-\sigma)^2$. Then

$$\tfrac{1}{2}\dot{W} = x\sigma(y-x) + y(rx - xz - y) + (z-r-\sigma)(xy - bz) = -\sigma x^2 - y^2 - bz^2 + b(r+\sigma)z,$$

again with all cross terms cancelling. With $c = (r+\sigma)/2$,

$$\dot{W} = -2\Big[\sigma x^2 + y^2 + b(z-c)^2\Big] + 2bc^2,$$

so $\dot{W} < 0$ outside the bounded ellipsoid $E_1 : \sigma x^2 + y^2 + b(z-c)^2 \le bc^2$. On E_1: $x^2 \le bc^2/\sigma$, $y^2 \le bc^2$, $0 \le z \le 2c$, so $W \le (b/\sigma + b + 4)c^2$ there. Choose $C > (b/\sigma + b + 4)c^2$; then the sphere $S = \{W \le C\}$ contains E_1, and the same compactness argument as in 9.2.2 shows that every trajectory enters S in finite time and stays. ■

9.2.4 *(z-axis is invariant)*

Solution. On $x = y = 0$ the vector field is $(0, 0, -bz)$. The function $(x, y, z)(t) = (0, 0, z_0 e^{-bt})$ satisfies the equations, and by uniqueness of solutions it is *the* trajectory through $(0, 0, z_0)$. So trajectories starting on the z-axis stay on it (and decay to the origin). ■

9.2.5 *(Stability diagram for $b = 1$)*

Solution. For $b = 1$: the origin is globally stable for $r < 1$ and loses stability at $r = 1$ in a pitchfork (creating $C^{\pm}$). The Hopf curve of $C^{\pm}$ is $r_H = \sigma(\sigma+4)/(\sigma-2)$, existing only for $\sigma > 2$; it has a vertical asymptote at $\sigma = 2$ and a minimum $r_H = 8 + 4\sqrt{3} \approx 14.9$ at $\sigma = 2 + 2\sqrt{3} \approx 5.46$. $C^{\pm}$ are stable for $1 < r < r_H$ (all $r > 1$ if $\sigma \le 2$) and unstable above the curve. ■

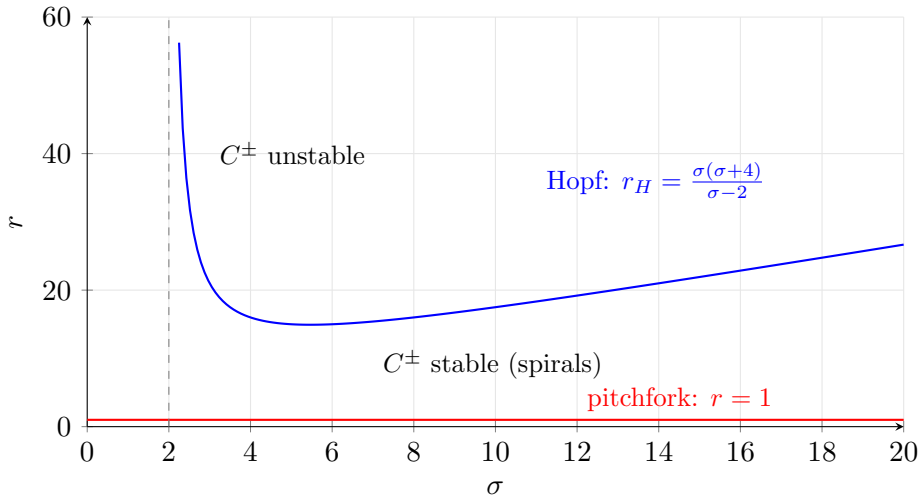


Figure 1: Partial stability diagram of the Lorenz system for $b = 1$. Below the red line ($r < 1$) the origin is globally stable.

9.2.6 *(Rikitake model of geomagnetic reversals)*

$\dot{x} = -\nu x + zy,\ \dot{y} = -\nu y + (z-a)x,\ \dot{z} = 1 - xy.$

Solution. **(a)** $\nabla \cdot f = -\nu - \nu + 0 = -2\nu < 0$: volumes contract like $e^{-2\nu t}$, so the system is dissipative.

(b) $\dot{z} = 0 \Rightarrow xy = 1$. $\dot{x} = 0 \Rightarrow z = \nu x/y = \nu x^2$, and $\dot{y} = 0 \Rightarrow z = a + \nu y/x = a + \nu/x^2$. Equating, $\nu(x^2 - x^{-2}) = a$. With $x^* = \pm k$:

$$x^* = \pm k, \quad y^* = \pm k^{-1}, \quad z^* = \nu k^2, \qquad \nu(k^2 - k^{-2}) = a.$$

(c) At $(k, k^{-1}, \nu k^2)$ we have $z^* - a = \nu k^{-2}$, so

$$J = \begin{pmatrix} -\nu & \nu k^2 & k^{-1} \\ \nu k^{-2} & -\nu & k \\ -k^{-1} & -k & 0 \end{pmatrix}, \qquad \det(J - \lambda I) = -\left[\lambda^3 + 2\nu\lambda^2 + s\lambda + 2\nu s\right] = -(\lambda + 2\nu)(\lambda^2 + s),$$

with $s = k^2 + k^{-2} > 0$. Thus

$$\lambda = -2\nu, \qquad \lambda = \pm i\sqrt{k^2 + k^{-2}}.$$

The same holds at $(-k, -k^{-1}, \nu k^2)$ by the symmetry $(x, y) \to (-x, -y)$. Both fixed points are non-hyperbolic: one stable direction and a pair of purely imaginary eigenvalues for *every* a, ν, so linearization cannot decide their stability (nonlinear terms on the 2D center manifold do). I did not carry out that center-manifold calculation. ■

3 Chaos on a Strange Attractor

9.3.1 *(Quasiperiodicity ≠ chaos)*

$\dot\theta_1 = \omega_1$, $\dot\theta_2 = \omega_2$, ω_1/ω_2 irrational.

Solution. **(a)** Solutions are $\theta_i(t) = \theta_i(0) + \omega_i t$ (on the torus). They never repeat, but there is no sensitive dependence on initial conditions: two trajectories θ, θ' satisfy $\theta(t) - \theta'(t) = \theta(0) - \theta'(0)$, so nearby points stay exactly as far apart forever. Long-term prediction is easy; "aperiodic" is not the same as chaotic.

(b) The separation $\delta(t)$ is constant, so $\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{|\delta(t)|}{|\delta(0)|} = 0$. All Liapunov exponents vanish; $\lambda_{\max} = 0$. ■

Numerical experiments, $\sigma = 10$, $b = 8/3$ (Exercises 9.3.2–9.3.7)

At $C^\pm$ the eigenvalues were computed numerically from the cubic of 9.2.1:

r	eigenvalues at $C^\pm$	consequence
10	$-12.5,\ -0.60 \pm 6.17i$	stable spiral
22	$-13.5,\ -0.086 \pm 9.11i$	stable spiral (weakly)
24.5	$-13.7,\ -0.0072 \pm 9.58i$	stable, barely ($r_H = 24.74$)
100	$-16.0,\ +1.16 \pm 18.1i$	unstable

9.3.2 *($r = 10$)*

Solution. All trajectories I tried converge to C^+ or C^-, spiralling in (oscillating x, y with decaying amplitude); which one depends on the sign side of the initial condition. No chaos. ■

9.3.3 *($r = 22$ (transient chaos))*

Solution. For some initial conditions the trajectory wanders chaotically on a butterfly-shaped set, switching irregularly between the two lobes, for a long time (here about 50 time units, and

$r = 10$, IC $(5, 5, 5)$: spiral into C^+; right panel shows the whole trajectory

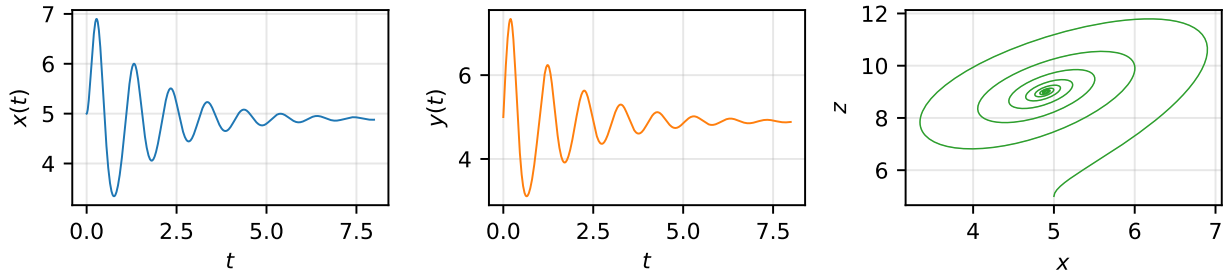


Figure 2: $r = 10$.

$r = 22$, IC $(-9.0, 10.6, 9.3)$: chaotic transient, then decay to a fixed point; right panel: whole trajectory

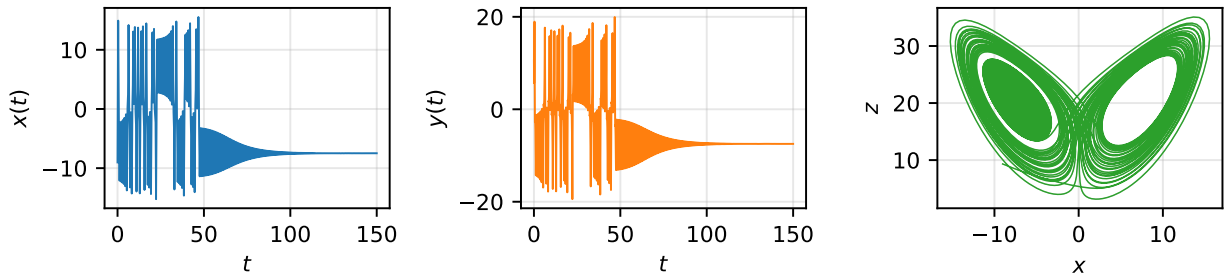


Figure 3: $r = 22$.

much longer for other starts), and then falls onto one of $C^{\pm}$ and spirals in. Chaos is transient: $C^{\pm}$ are the only attractors. ■

9.3.4 *($r = 24.5$ (chaos and stable point coexist))*

Solution. Here $C^{\pm}$ are still stable ($r < r_H \approx 24.74$) but a strange attractor exists too (it appears for $r \gtrsim 24.06$). The outcome depends on the initial condition: some starts give sustained chaos (checked up to $t = 300$), others converge to $C^{\pm}$ (very slowly, since $\operatorname{Re}\lambda \approx -0.007$). ■

9.3.5 *($r = 100$ (surprise))*

Solution. The surprise is that at this large r the motion is *not* chaotic in the long run. From several initial conditions I found a chaotic transient of about 200 time units followed by convergence to a stable periodic orbit whose z-maxima repeat in a cycle of three values ($\approx 114.6, 123.5, 144.4$). The limit cycle is a periodic window inside the chaotic regime. ■

9.3.6 *($r = 126.52$)*

Solution. Again a long chaotic transient (a couple of hundred time units) followed by a periodic orbit. Two different initial conditions gave the same repeating sequence of six z-maxima ($\approx 147, 163, 157, 168, 151, 177$), so the attractor I observe is a stable limit cycle, with a more complicated loop structure than at $r = 100$. ■

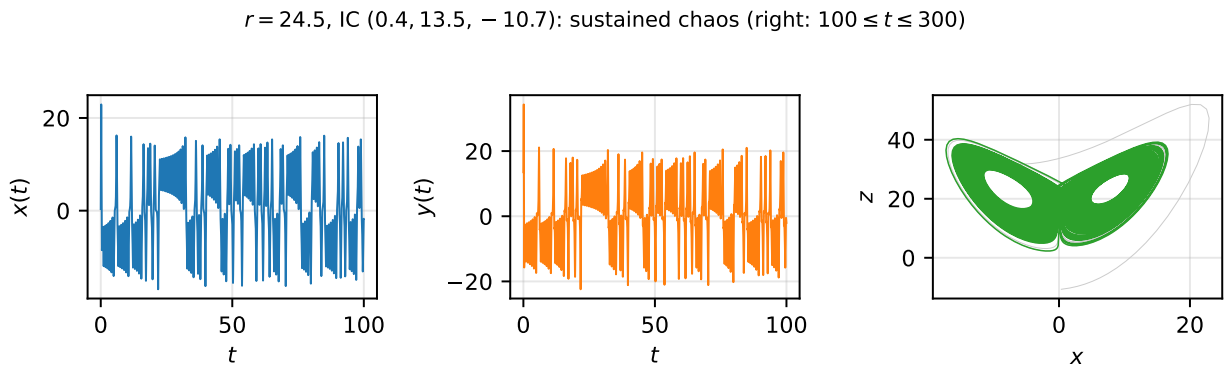


Figure 4: $r = 24.5$: sustained chaos.

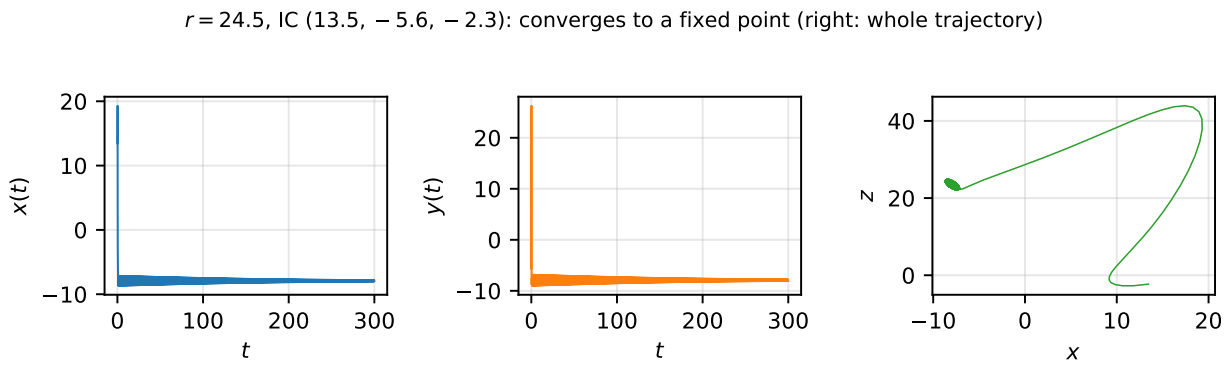


Figure 5: $r = 24.5$: convergence to a fixed point.

9.3.7 *($r = 400$)*

Solution. After a short transient the trajectory settles on a simple limit cycle: x oscillates almost periodically between roughly $\pm$ 50–60, with a fast period (≈ 0.35 time units). This is consistent with the known fact that for large r the Lorenz system has a globally attracting periodic orbit. (That fact is from the literature, not derived here; see also 9.5.5.) ■

9.3.8 *(Practice with the definition of an attractor)*

$\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$, $D = \{x^2 + y^2 \leq 1\}$.

Solution. **(a)** Yes. $r = 0$ and $r = 1$ are fixed in r; for $0 < r < 1$, $\dot{r} > 0$ so r increases toward 1 forward in time, and decreases to 0 backward in time. So trajectories starting in D stay in D for all t.

(b) Yes. Every point with $r_0 > 0$ (in particular all points in a neighborhood such as $r < 2$) has $r(t) \to 1$, hence its distance to D tends to 0.

(c) No. An attractor must be *minimal*: no proper closed subset may itself be invariant and attracting. D contains the unit circle, which is invariant and attracts the same open set. So D is not minimal, hence not an attractor.

(d) The unit circle $r = 1$ is a closed invariant set; every orbit $r_0 > 0$ is attracted to it, so it attracts

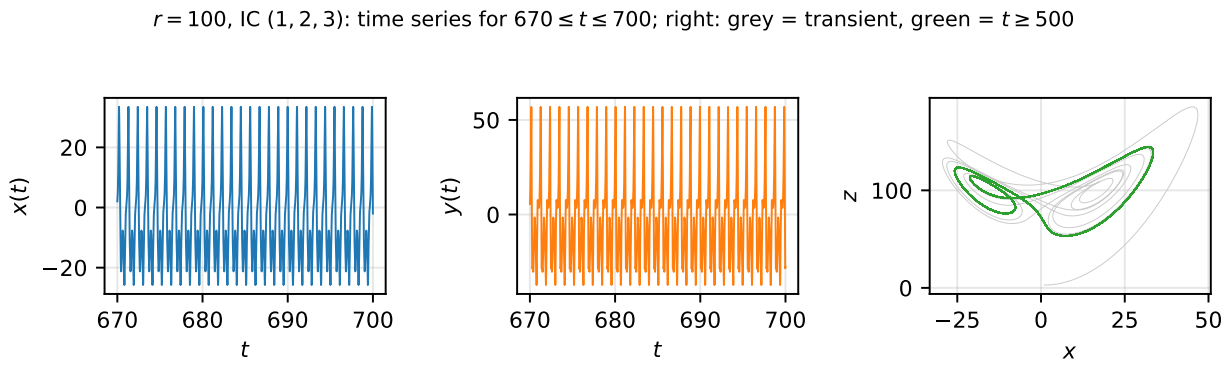


Figure 6: $r = 100$.

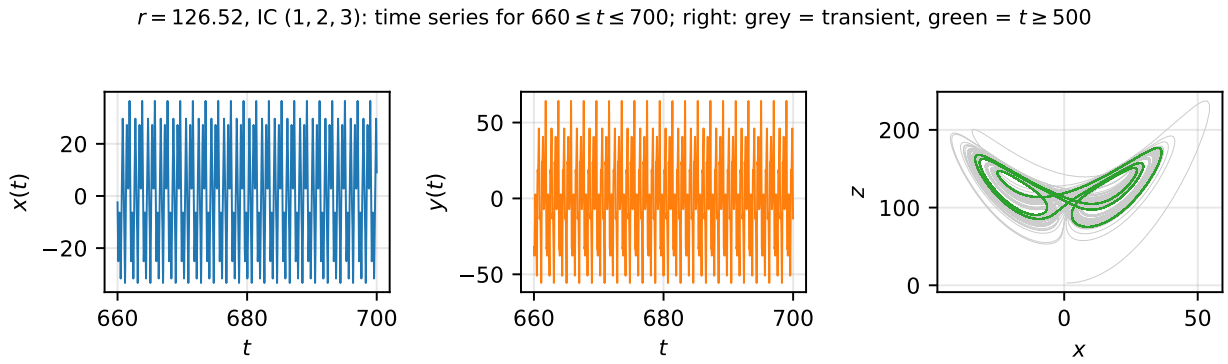


Figure 7: $r = 126.52$.

an open set; it is a single periodic orbit (θ increases at unit rate), so any nonempty closed invariant subset is all of it: minimal. Hence it is an attractor, with basin of attraction $\{r > 0\} = \mathbb{R}^2 \setminus \{0\}$. ■

9.3.9 *(Exponential divergence)*

Estimate the largest Liapunov exponent for $r = 28$, $\sigma = 10$, $b = 8/3$.

Solution. I integrated two trajectories on the attractor starting 10^{-8} apart and plotted $\ln \|\delta(t)\|$ (Fig. 9): after a brief adjustment the separation grows exponentially until it saturates at the size of the attractor. Fitting the straight part for four random directions gave slopes 0.97–1.07 (finite-time values scatter because the local expansion rate varies around the attractor). A more reliable estimate uses repeated renormalization of the separation (Benettin's method) over a long run ($T = 1500$):

$$\boxed{\lambda_1 \approx 0.90},$$

consistent with the accepted value ≈ 0.906. ■

9.3.10 *(Time horizon)*

Solution. Two trajectories were started at distance 10^{-5} along x and $x(t)$ plotted for both (Fig. 10): they are indistinguishable at first, then separate completely near $t \approx 13$ and from then on are unrelated. The time for the separation to reach size 1 for different initial gaps was 2.9

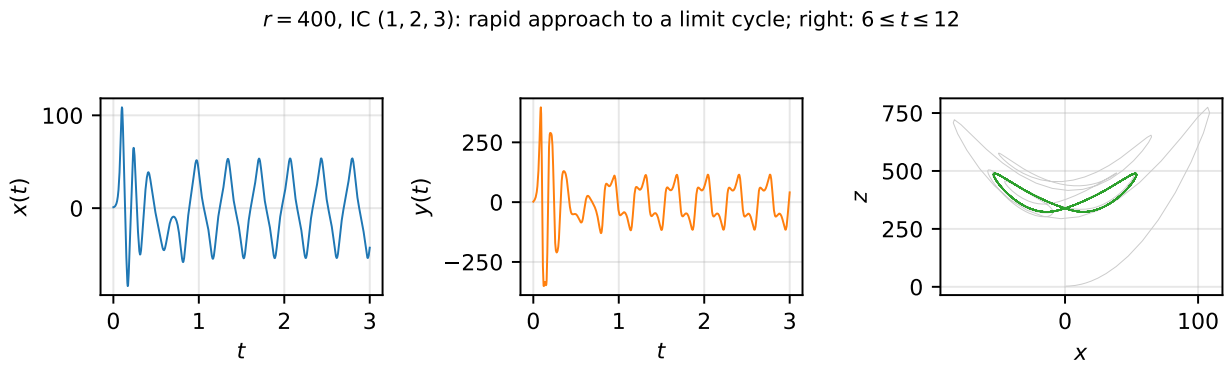


Figure 8: $r = 400$.

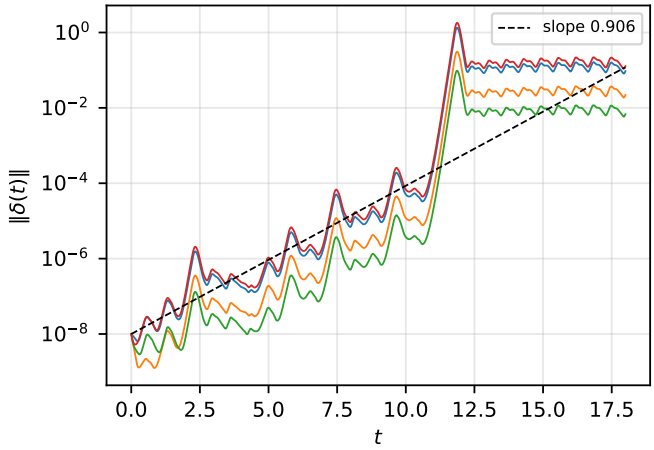


Figure 9: Separation of four pairs of nearby trajectories (log scale); the dashed line has slope 0.906.

(10^{-3}), 12.7 (10^{-5}), 16.6 (10^{-7}), 21.1 (10^{-9}): every factor 100 gained in precision buys only about 4–4.5 extra time units, roughly $\ln(100)/\lambda_1 \approx 5$. This is the "time horizon" $t_{\text{horizon}} \sim \frac{1}{\lambda_1}\ln(a/\delta_0)$ (a = tolerable error): prediction time grows only logarithmically with accuracy. ■

4 Lorenz Map

9.4.1 *(Computer work)*

Solution. For $r = 28$, $\sigma = 10$, $b = 8/3$ I integrated for $T = 2000$ and recorded the successive local maxima z_n of $z(t)$ (2594 maxima after discarding transients), then plotted z_{n+1} against z_n (parabolic interpolation was used to locate each maximum accurately). The points fall on an almost one-dimensional curve: a tent-like map with a single peak near $z_n \approx 38.5$ where $z_{n+1} \approx 46.5$. The maxima lie in [30.9, 46.5]; the curve crosses the diagonal at $z \approx 39.8$, and in my data its slope has modulus greater than about 1 almost everywhere, so the fixed point is unstable and nearby orbits diverge, as expected of chaos. ■

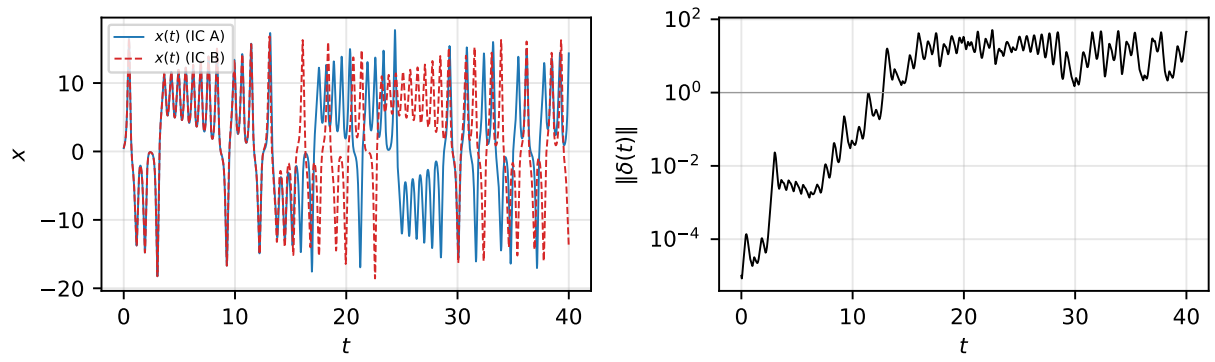


Figure 10: Left: $x(t)$ for two nearly identical starts. Right: their separation.

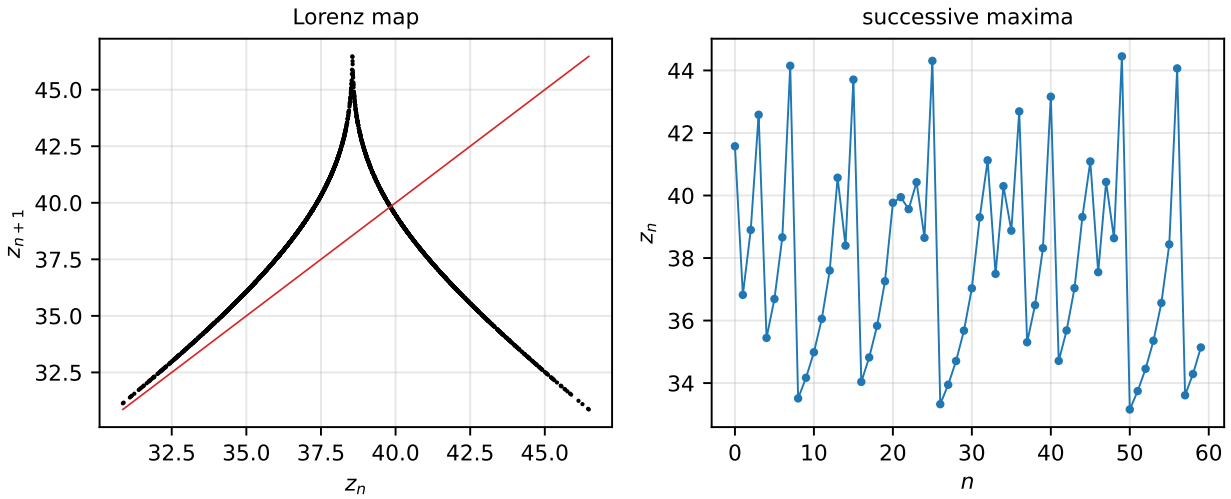
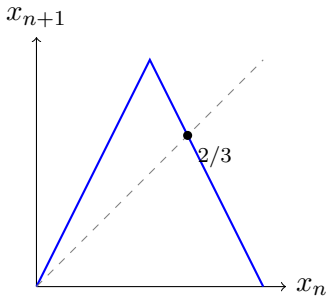


Figure 11: Lorenz map (left, diagonal in red) and the first 60 maxima (right).

9.4.2 *(Tent map as a model of the Lorenz map)*

$x_{n+1} = 2x_n$ for $0 \le x_n \le \frac{1}{2}$, $x_{n+1} = 2 - 2x_n$ for $\frac{1}{2} \le x_n \le 1$.

Solution. **(a)** Its graph rises linearly from $(0,0)$ to $(\frac{1}{2},1)$ and falls linearly to $(1,0)$: a tent (triangle).



(b) Fixed points: $x = 2x \Rightarrow x = 0$; $x = 2 - 2x \Rightarrow x = 2/3$. Since $|f'| = 2 > 1$ everywhere, both are **unstable**.

(c) $\{2/5, 4/5\}$: $f(2/5) = 4/5$, $f(4/5) = 2 - 8/5 = 2/5$. Its multiplier is $f'(2/5)f'(4/5) = 2 \cdot (-2) = -4$, $|-4| > 1$: unstable.

(d) Period 3: $\{2/7, 4/7, 6/7\}$ ($2/7 \to 4/7 \to 6/7 \to 2/7$) with multiplier $2 \cdot (-2)(-2) = 8$, and $\{2/9, 4/9, 8/9\}$ with multiplier $2 \cdot 2 \cdot (-2) = -8$. Period 4: $\{2/15, 4/15, 8/15, 14/15\}$ (multiplier 16), $\{2/17, 4/17, 8/17, 16/17\}$ (multiplier -16), and $\{6/17, 12/17, 10/17, 14/17\}$ (multiplier -16). In general f^n is piecewise linear with 2^n branches of slope $\pm 2^n$, each crossing the diagonal once, so there are 2^n points with $f^n(x) = x$ (giving $2^3 - 2 = 6$ points = two 3-cycles and $2^4 - 2^2 = 12$ points = three 4-cycles), and every periodic orbit of period n has multiplier $\pm 2^n$: **all are unstable**. ■

5 Exploring Parameter Space

9.5.1 *($r = 166.3$ (intermittent chaos))*

Solution. The attractor is a thick, irregular version of a simple loop (middle panel). The sequence of z-maxima stays in a band ($\approx$ 195–230) with occasional large excursions, i.e. the motion is aperiodic. I could not resolve clean laminar periodic stretches on the time scale I integrated, so I only report the irregularity here; the intermittency scenario itself (near-periodic motion interrupted by bursts, near a saddle-node at $r \approx 166.07$) is from the literature, not something I verified. ■

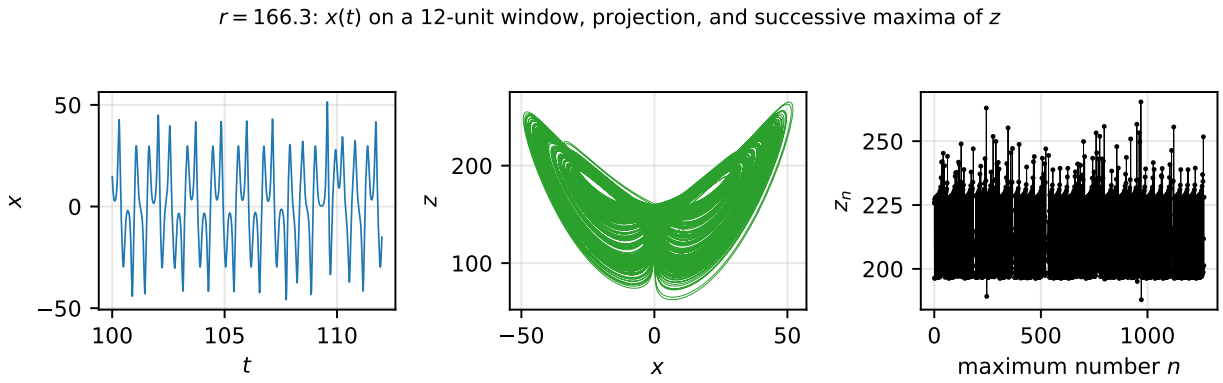


Figure 12: $r = 166.3$.

9.5.2 *($r = 212$ (noisy periodicity))*

Solution. $x(t)$ oscillates with an almost constant period ($\approx$ 0.35 time units), and the (x, z) projection is a closed loop of finite thickness: a periodic orbit with small irregular jitter. The maxima of z are scattered in a narrow band ($\approx$ 253–293), so it is not exactly periodic; a crude Liapunov estimate from a short run was positive. ■

9.5.3 *($145 < r < 166$ (period doubling))*

Solution. I plotted the local maxima of $z(t)$ for $r \in [145, 166]$ in steps of 0.3 (figure below). For $r \gtrsim 155$ there are two distinct maxima per cycle (one per lobe: the basic cycle). Near $r \approx 154.5$

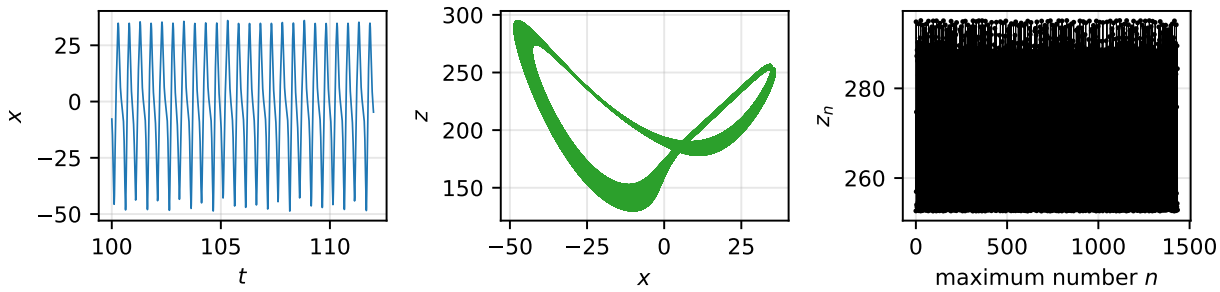


Figure 13: $r = 212$.

this splits into four maxima, then eight near $r \approx 148$, and below $r \approx 147$ there are many distinct maxima: a period-doubling cascade to chaos as r decreases. The resolution (step 0.3) is too coarse to locate the bifurcation values accurately, and further doublings are not resolved. ■

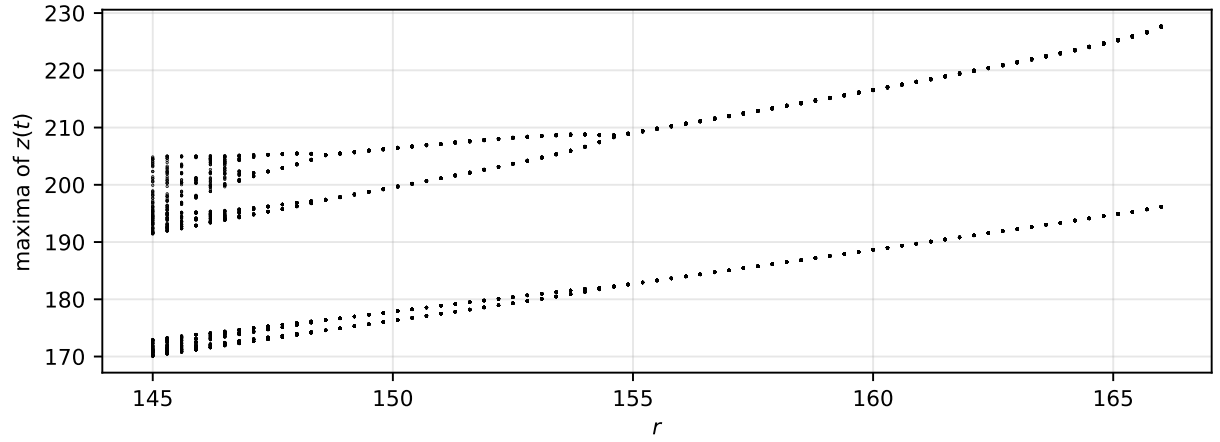


Figure 14: Maxima of $z(t)$ versus r.

9.5.4 *(Hysteresis between a fixed point and a strange attractor)*

$r = 24.4 + \sin \omega t$.

Solution. Fixed points are stable for $r < 24.74$ and the strange attractor exists for $r > 24.06$, so $r \in [23.4, 25.4]$ sweeps through both thresholds. With $\omega = 0.004$ and a start at C^+ (figure below), x stays near the fixed point until r is well above 24.74 (the instability is very weak at first, so chaos only shows up near $t \approx 300$ and $r \approx 25.3$), then becomes chaotic. As r falls back through 24.74 and even through 24.06 down to 23.4, the motion stays chaotic for the whole run ($t \leq 1960$): the system does not return to the fixed point at the same r where it left. That is the hysteresis. (At $r < 24.06$ chaos is only transient, but its lifetime is longer than the time spent there. With the faster $\omega = 0.01$ a start in the chaotic state remained chaotic throughout.) ■

9.5.5 *(Lorenz equations for large r)*

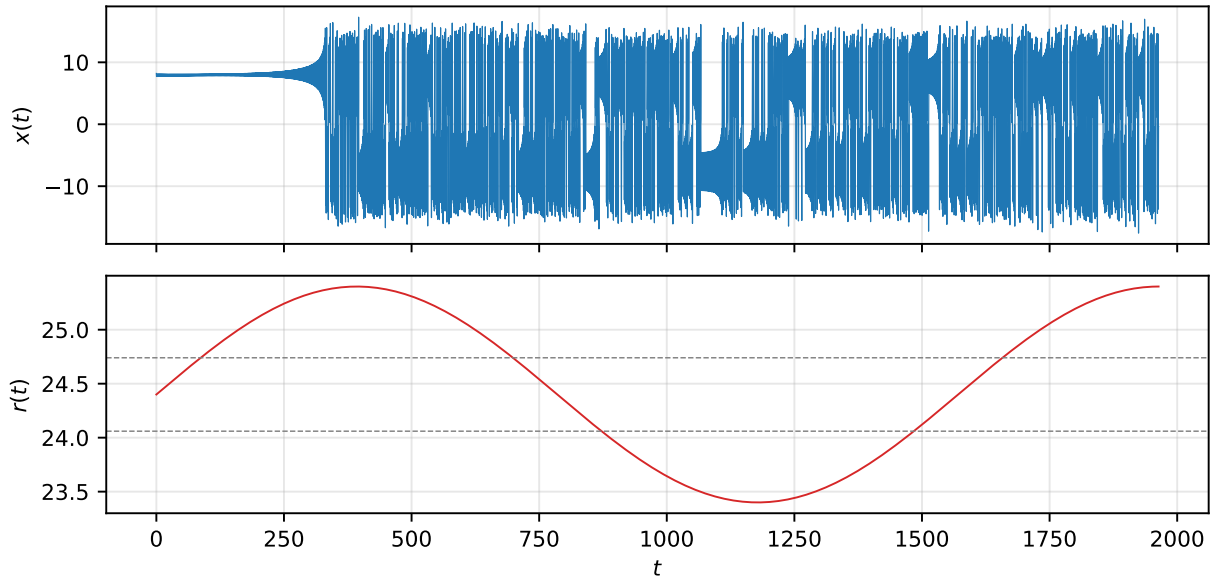


Figure 15: Top: $x(t)$. Bottom: $r(t)$, with the thresholds 24.06 and 24.74 dashed.

Solution. **(a)** Let $\varepsilon = r^{-1/2}$ and

$$x = \frac{X}{\varepsilon}, \qquad y = \frac{Y}{\sigma\varepsilon^2}, \qquad z = \frac{1}{\varepsilon^2}\Big(1 + \frac{Z}{\sigma}\Big) = r + \frac{Z}{\sigma\varepsilon^2}, \qquad t = \varepsilon\tau.$$

Substituting (with $' = \mathrm{d}/\mathrm{d}\tau$):

$$X' = Y - \sigma\varepsilon X, \qquad Y' = -XZ - \varepsilon Y, \qquad Z' = XY - b\varepsilon(\sigma + Z).$$

As $\varepsilon \to 0$ these become $X' = Y$, $Y' = -XZ$, $Z' = XY$.

(b) $\frac{\mathrm{d}}{\mathrm{d}\tau}(Y^2 + Z^2) = 2Y(-XZ) + 2Z(XY) = 0$ and $\frac{\mathrm{d}}{\mathrm{d}\tau}(X^2 - 2Z) = 2XY - 2XY = 0$. So $Y^2 + Z^2$ and $X^2 - 2Z$ are conserved.

(c) $\nabla \cdot (Y, -XZ, XY) = \partial_X Y + \partial_Y(-XZ) + \partial_Z(XY) = 0$, so by Liouville's theorem the flow preserves volume.

(d) For large r (very strong pumping q_1 in the waterwheel) the gravitational/inertial effects dominate while leakage and rotational damping, the dissipative terms (of relative size $O(\varepsilon)$), are comparatively negligible, so the leading-order dynamics is like a frictionless inverted pendulum: conservative.

(e) Numerically (figure below) the limit system gives *periodic* motion for the three initial conditions I tried: X oscillates through both signs (the wheel rocks back and forth), and both conserved quantities are constant to 10^{-10}. Because the system is conservative, each initial condition lies on its own periodic orbit rather than there being one attracting cycle. This agrees qualitatively with the periodic motion seen at large r in the Lorenz system (9.3.7), and the small $O(\varepsilon)$ dissipative terms are what would select one particular limit cycle out of this family (not checked here). ■

9.5.6 *(Transient chaos)*

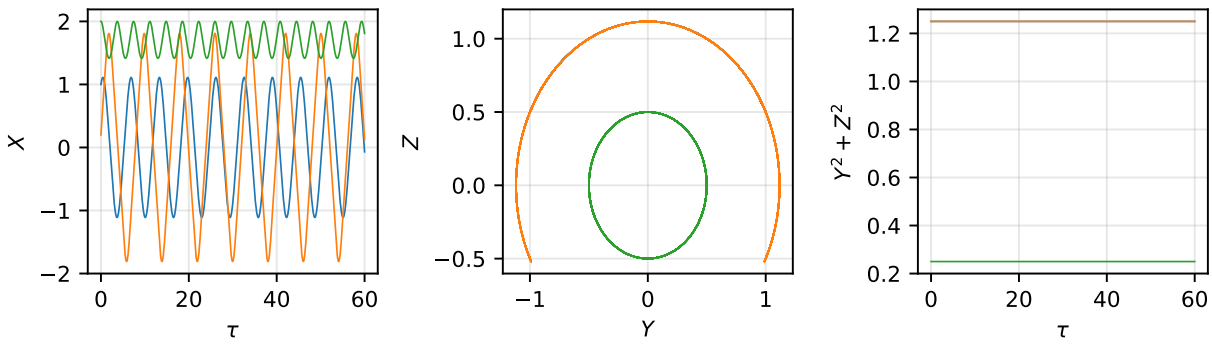


Figure 16: Limit system $X' = Y$, $Y' = -XZ$, $Z' = XY$ for three initial conditions. Left: $X(\tau)$; middle: (Y, Z) plane; right: $Y^2 + Z^2$ is constant.

Solution. For $r = 21$, $\sigma = 10$, $b = 8/3$ I integrated many initial conditions and counted sign changes of x before the trajectory settles on $C^{\pm}$ (figure below). Transient chaos (many switches between lobes): $(10.5, -14.8, 19.7)$, $(-14.8, -16.7, 21.3)$, $(-2.8, 0.8, 23.0)$. No transient chaos (straight into the nearest fixed point, no sign changes): $(-11.0, -8.0, 19.9)$, $(15.0, 6.5, 19.7)$, $(4.2, 5.5, 12.1)$.

Rule of thumb. On a 31×31 grid in the plane $y_0 = 0$, $0 \le z_0 \le 50$, $|x_0| \le 25$, about 60–75% of starts with $|x_0| < 6$ (near the z-axis, which is part of the origin's stable manifold) gave transient chaos, versus only about 20% for $|x_0| > 15$; starts far from the z-axis typically fall straight onto the fixed point on their own side. The dividing set is intricate rather than a clean boundary, so this is only a tendency: starts near the z-axis and inside the "butterfly" region are more likely to be transient chaos; starts far out are more likely to go directly to $C^{\pm}$. ■

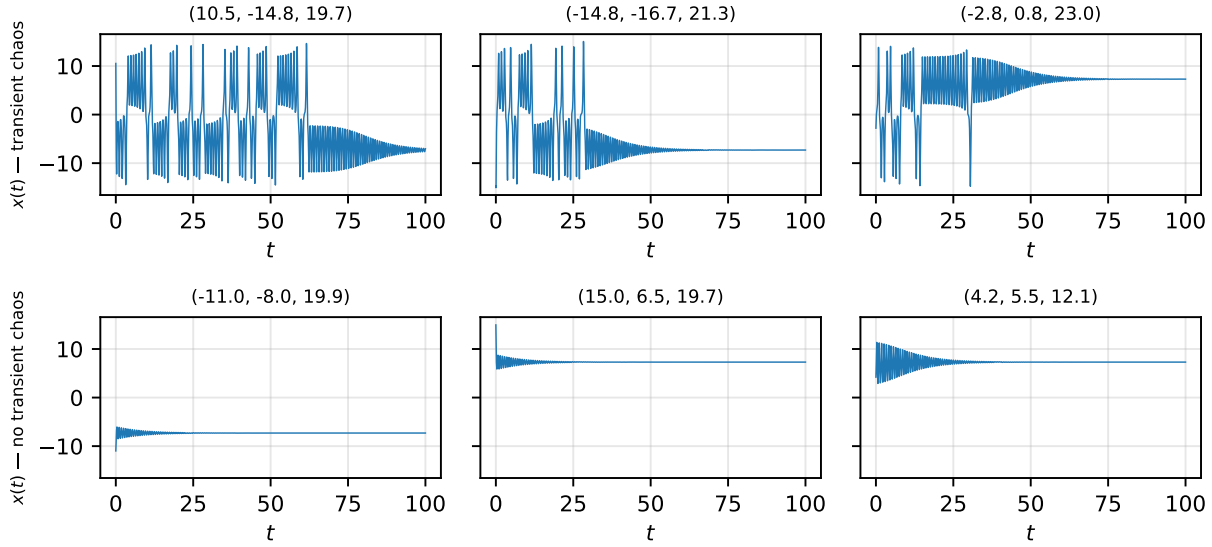


Figure 17: $x(t)$ at $r = 21$. Top: transient chaos. Bottom: no transient chaos. Titles give the initial condition.

6 Using Chaos to Send Secret Messages

In scaled variables (book): driver $\dot{u} = \sigma(v-u)$, $\dot{v} = ru - v - 20uw$, $\dot{w} = 5uv - bw$; receiver the same with $u \to u(t)$ in the v_r, w_r equations; error $\mathbf{e} = (u-u_r,\, v-v_r,\, w-w_r)$ obeys $\dot{e}_1 = \sigma(e_2 - e_1)$, $\dot{e}_2 = -e_2 - 20u(t)e_3$, $\dot{e}_3 = 5u(t)e_2 - be_3$.

9.6.1 *(Exponentially fast synchronization)*

Solution. **(a)** With $V = \frac{1}{2}e_2^2 + 2e_3^2$:

$$\dot{V} = e_2(-e_2 - 20ue_3) + 4e_3(5ue_2 - be_3) = -e_2^2 - 4be_3^2,$$

the chaotic terms $\mp 20ue_2e_3$ cancel. Since $-e_2^2 - 4be_3^2 \le -kV$ means $k(\frac{1}{2}e_2^2 + 2e_3^2) \le e_2^2 + 4be_3^2$, it holds with $k = \min(2, 2b)$ (for $b = 8/3$, $k = 2$). Thus $\dot{V} \le -kV$.

(b) By Gronwall, $V(t) \le V(0)e^{-kt}$, and $e_2^2 \le 2V$, $e_3^2 \le V/2$, so $|e_2|, |e_3| \le Ce^{-kt/2}$.

(c) $\dot{e}_1 = -\sigma e_1 + \sigma e_2$, so $e_1(t) = e_1(0)e^{-\sigma t} + \sigma \int_0^t e^{-\sigma(t-s)} e_2(s)\,\mathrm{d}s$. With $|e_2(s)| \le Ce^{-ks/2}$, the integral is $O\big(e^{-\min(\sigma, k/2)t}\big)$ (times t if $\sigma = k/2$). So $e_1 \to 0$ exponentially as well. ■

9.6.2 *(Pecora and Carroll's approach)*

Receiver: $x_r = x$, $\dot{y}_r = rx - y_r - xz_r$, $\dot{z}_r = xy_r - bz_r$.

Solution. **(a)** $e_1 = x - x_r \equiv 0$ by construction. Subtracting the receiver equations from the driver equations,

$$\dot{e}_2 = (rx - y - xz) - (rx - y_r - xz_r) = -e_2 - xe_3, \qquad \dot{e}_3 = (xy - bz) - (xy_r - bz_r) = xe_2 - be_3.$$

(b) For $V = e_2^2 + e_3^2$ (positive definite),

$$\dot{V} = 2e_2(-e_2 - xe_3) + 2e_3(xe_2 - be_3) = -2e_2^2 - 2be_3^2 < 0 \quad \text{unless } e_2 = e_3 = 0,$$

since the $x(t)$ terms cancel exactly. So V is a Liapunov function.

(c) $e_2, e_3 \to 0$ for every initial error, whatever the (chaotic) $x(t)$ is; in fact $\dot{V} \le -2\min(1, b)V$, so exponentially. Hence $y_r \to y$, $z_r \to z$: the receiver synchronizes with the transmitter. ■

9.6.3 *(Computer experiments on synchronized chaos)*

$r = 60$, $\sigma = 10$, $b = 8/3$.

Solution. Driver started at $(1, 2, 20)$, receiver at $(y_r, z_r) = (-8, 5)$. (a) y and y_r merge within a few time units; the error norm falls from $O(10)$ to about 10^{-13} by $t = 20$ (it decays exponentially, as proved above). (b) After the transient the (y, z) projections of the two systems lie on top of each other on the same strange attractor. ■

9.6.4 *(Some drives don't work)*

Solution. **(a)** With $z(t)$ as drive the receiver is $\dot{x}_r = \sigma(y_r - x_r)$, $\dot{y}_r = rx_r - y_r - x_r z(t)$. The error equations are $\dot{e}_1 = \sigma(e_2 - e_1)$, $\dot{e}_2 = (r - z(t))e_1 - e_2$ ($e_1 = x - x_r$, $e_2 = y - y_r$): a linear system with a chaotic coefficient and no obvious Liapunov function. Numerically ($r = 60$) the receiver does *not* synchronize: after 40 time units the error is still $O(1)$ (error norm 0.34 at $t = 5$, 3.5 at $t = 10$, 0.39

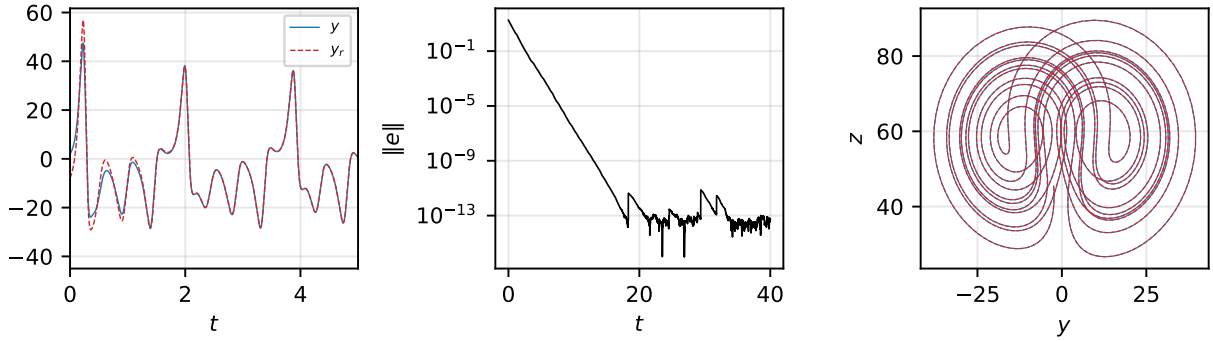


Figure 18: x-drive. Left: y and y_r. Middle: error norm (log scale). Right: (y, z) projections (indistinguishable).

at $t = 40$), and x_r is only poorly correlated with x. A Lyapunov-exponent estimate for this error system (the conditional exponent) came out at ≈ -0.007, indistinguishable from 0 in my run, i.e. at best extremely marginal, which is consistent with the lack of synchronization on any practical time scale.

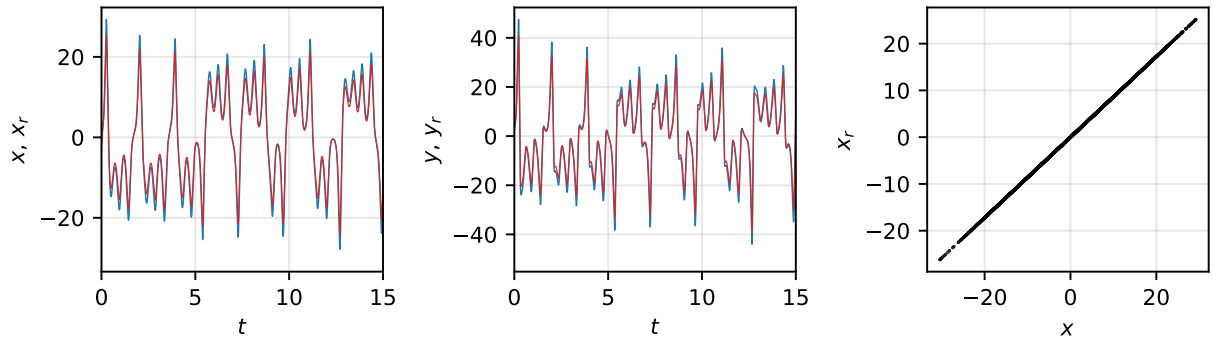


Figure 19: z-drive. Left/middle: x, x_r and y, y_r (blue: driver, red: receiver). Right: x_r against x.

(b) y *does* work as a drive. The receiver is $\dot{x}_r = \sigma(y - x_r)$, $\dot{z}_r = x_r y - b z_r$, and the error equations are

$$\dot{e}_1 = -\sigma e_1, \qquad \dot{e}_3 = y(t)\, e_1 - b e_3,$$

a triangular system: $e_1 \to 0$ exponentially, and then e_3 is a stable linear system forced by a vanishing input, so $e_3 \to 0$. Numerically the error was 4.6×10^{-5} at $t = 5$ and $\sim 10^{-14}$ at $t = 20$. ■

9.6.5 *(Masking)*

Solution. Transmit $s(t) = x(t) + m(t)$ with $m = A \sin \omega t$, $A = 0.1$ ($|x|$ is of order 20–40), $r = 60$. The receiver is driven by s; the recovered message is $\hat{m} = s - x_r$. Since $x - x_r = \hat{m} - m$, the error is exactly how far x_r is from x. The relative RMS error $\|\hat{m} - m\|/\|m\|$ (after transients, $t > 20$):

ω	0.1	0.3	1	3	10	30	100
relative error	2.6	3.6	3.3	2.4	2.0	0.32	0.015

So for low frequencies the recovered message is *worse than useless* (error larger than the message): the receiver itself responds to m because m is injected into the $\dot{y}_r$ and $\dot{z}_r$ equations with the large chaotic coefficients $(r - z_r)$ and y_r, giving an error of order m times $O(1)$. For high frequency ($\omega \gg \sigma = 10$) the receiver's x_r equation acts like a low-pass filter ($\dot{x}_r = \sigma(y_r - x_r)$), x_r ignores the rapid wiggles, and $\hat{m} \approx m$ with error falling quickly. So with these parameters the scheme works well only for messages whose frequency is well above σ (here $\omega \gtrsim 30$). ■

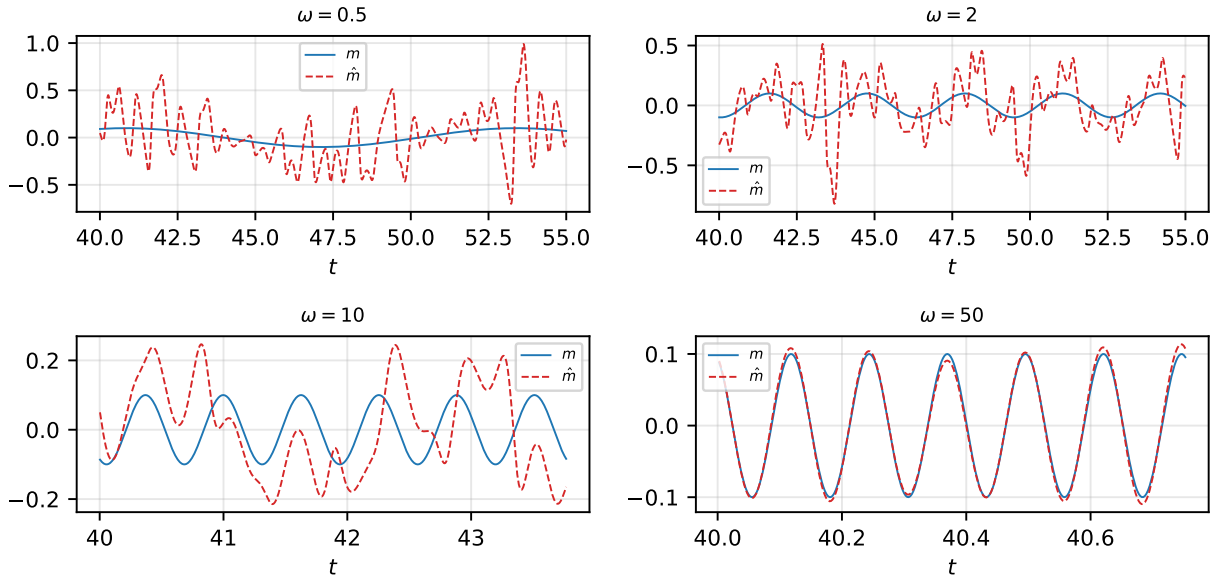


Figure 20: Message m (blue) and recovered $\hat{m}$ (red dashed) for $\omega = 0.5, 2, 10, 50$.

9.6.6 *(Lorenz circuit)*

Derive the equations of the transmitter circuit of Figure 9.6.1.

Solution. *Assumptions.* Ideal op-amps; multipliers with output = $(\text{input}_1 \times \text{input}_2)/10\,\text{V}$ (standard); matched resistors where noted. I read the wiring from the figure in the book; I could not check it against the original paper, so the grouping below is my reading of the diagram.

Stage 1 (u). Op-amp 1 with R_1, R_2, R_3, R_4 is a difference amplifier with inputs v (to R_1, inverting) and u (to R_2, non-inverting). With $R_3/R_2 = R_4/R_1$, $V_1 = \frac{R_4}{R_1}(u - v)$. Op-amp 2 with R_5 and C_1 is an inverting integrator, $u = -\frac{1}{R_5 C_1}\int V_1\,\mathrm{d}t$, so

$$\dot{u} = \frac{R_4}{R_1 R_5 C_1}(v - u).$$

Stage 2 (v). Op-amp 3 (R_6, R_7) is a non-inverting amplifier: output $(1 + R_7/R_6)u$. Its output goes through the divider R_{10}, R_{11} to the non-inverting input of op-amp 4; the inverting input receives v through R_8 and the upper multiplier output $uw/10$ through R_9; the feedback resistor is R_{12}. With

$\alpha = \frac{R_{11}}{R_{10}+R_{11}}$,

$$V_4 = \Big(1 + R_{12}\big(\tfrac{1}{R_8} + \tfrac{1}{R_9}\big)\Big)\alpha\Big(1 + \tfrac{R_7}{R_6}\Big)u - \frac{R_{12}}{R_8}v - \frac{R_{12}}{R_9}\frac{uw}{10}.$$

Op-amp 5 (R_{13}, R_{14}) is an inverting amplifier, $V_5 = -\frac{R_{14}}{R_{13}}V_4$, and op-amp 6 ($R_{15}, C_2$) an inverting integrator, $v = -\frac{1}{R_{15}C_2}\int V_5\,\mathrm{d}t$. Hence

$$\dot v = \frac{R_{14}}{R_{13}R_{15}C_2}\,V_4.$$

Stage 3 (w). The lower multiplier forms $uv/10$. Op-amp 7 ($R_{16}, R_{17}, R_{18}, R_{19}$) has the multiplier output on its inverting input (through R_{16}) and a fraction $\beta = \frac{R_{18}}{R_{17}+R_{18}}$ of w on its non-inverting input: $V_7 = -\frac{R_{19}}{R_{16}}\frac{uv}{10} + \big(1 + \frac{R_{19}}{R_{16}}\big)\beta w$. Op-amp 8 ($R_{20}, C_3$) integrates inverting: $\dot w = -V_7/(R_{20}C_3)$, so

$$\dot w = \frac{R_{19}}{10\,R_{16}R_{20}C_3}\,uv - \frac{(1 + R_{19}/R_{16})\,\beta}{R_{20}C_3}\,w.$$

Result. The circuit obeys $\dot u = \sigma'(v-u)$, $\dot v = r'u - \mu v - \eta\,uw$, $\dot w = \xi\,uv - b'w$ with $\sigma' = \frac{R_4}{R_1R_5C_1}$, $\mu = \frac{R_{14}R_{12}}{R_{13}R_{15}C_2R_8}$, $\eta = \frac{R_{14}R_{12}}{10R_{13}R_{15}C_2R_9}$, $\xi = \frac{R_{19}}{10R_{16}R_{20}C_3}$, $b' = \frac{(1+R_{19}/R_{16})\beta}{R_{20}C_3}$ and r' the first-term coefficient above (adjustable through R_{11} and R_{18}; R_5 is also variable). Measuring time in units of $1/\mu$ (i.e. $t' = \mu t$) gives the book's equation (1),

$$\dot u = \sigma(v-u), \quad \dot v = ru - v - 20uw, \quad \dot w = 5uv - bw,$$

with $\sigma = \sigma'/\mu$, $r = r'/\mu$, $20 = \eta/\mu$, $5 = \xi/\mu$, $b = b'/\mu$, which can be arranged by choosing the resistor and capacitor values (the book does not list them). ■